

WIP: Workshop Activities Focused on Hands-On Abilities For Engineering Students

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Abstract

This theory work-in-progress paper presents a two-part, hands-on workshop designed to address the gap between engineering students' theoretical knowledge and practical skills. Despite students' technological proficiency, many lack direct experience with industrial equipment and essential workplace skills, as highlighted by employer feedback. The workshop engages undergraduate engineering students in the design and fabrication of custom printed circuit boards (PCBs) and 3D-printed enclosure cases, utilizing industry-standard tools such as Fusion 360 and SolidWorks. The process includes schematic creation, PCB board layout, component selection, and 3D modeling, followed by manufacturing, assembly, and functional testing of the finished products. The initiative aims to enhance students' technical competencies, teamwork, and sense of belonging within the engineering community. Preliminary outcomes, supported by pre- and post-workshop surveys, suggest improved hands-on skills and increased student engagement.

1 Introduction

Undergraduate engineering education often struggles to balance theoretical instruction with practical, hands-on experience. While students become proficient with software tools and theoretical concepts, research and industry feedback highlight a significant gap between classroom learning and the skills required in professional engineering practice^{1,2}. Employers frequently note that recent graduates lack experience working directly with engineering equipment, manufacturing processes, and collaborative problem-solving in realistic settings.

To address this gap, many engineering programs are incorporating experiential learning opportunities, such as project-based courses, makerspaces, and service-learning initiatives. These approaches engage students in the full engineering design process—from conceptualization and computer-aided design (CAD) to fabrication, assembly, and troubleshooting³. For example, the Engineering Projects in Community Service (EPICS) program has demonstrated that engaging students in real-world projects not only builds technical proficiency but also strengthens essential skills such as teamwork, communication, and adaptability⁴.

In a similar effort, we developed a two-part, hands-on workshop that exemplifies these educational innovations in this paper. In this workshop, students design and fabricate custom printed circuit boards (PCBs) and 3D-printed cases, using industry-standard tools such as Autodesk Fusion 360 and SolidWorks. Through this process, they gain practical experience with electronics, mechanical design, and programming. The workshop structure—spanning initial design through assembly and testing—mirrors real engineering workflows and provides students with opportunities to develop both technical and interpersonal skills. This experiential approach not only provides valuable technical practice but also fosters collaboration and a sense of accomplishment. In this research paper, we outline the workshop structure, share outcomes from previous participants, and discuss plans for assessing student learning through pre- and post-workshop surveys.

2 Workshop Activities

2.1 PCB Design

In the PCB design process, students start by understanding the project's goals and what electronic components they need. Next, they create libraries for all the components. They can do this by checking manufacturers' datasheets or by downloading existing libraries from websites like snapeda.com or ultralibrarian.com. Once the libraries are ready, students use Fusion 360 to create the project schematic. This involves placing each component in the schematic editor and connecting them according to the circuit's logic, ensuring all electrical connections are accurate. After completing the schematic, the next step is to design the board layout in Fusion 360. The software helps arrange each component on the board and allows for manual routing of the copper traces that connect them. It's important to optimize for signal quality and ease of manufacturing. Care must be taken to avoid crossing traces and to meet design rules about clearances and trace widths. Once the layout is finished, Fusion 360 generates the necessary Gerber files. These files are standard formats that PCB manufacturers need to create the board.

2.2 PCB Enclosure Design

During this activity, students measure the size of the PCB, the 9V battery, and the ESP8266 module to find the minimum space needed inside an enclosure. They use 3D modeling software like SolidWorks to create a model that starts with a base to securely hold all components while keeping the size small. They place mounting features, such as standoffs and slots, to keep the PCB and battery in place and prevent movement during use. They add openings and cutouts for connectors, switches, and ventilation to ensure easy access and proper airflow for the ESP8266. The design is also refined to include snap fits, screw holes, or other fasteners for easy assembly and maintenance. After designing the 3D case, the student will print it in the Library makerspace before the next workshop. This ensures the enclosure is ready for the next phase of the project. The result is a custom case that fits all components neatly and provides solid protection.

2.3 Product Assembly

Once the designed PCB is returned from the manufacturer and the 3D-printed enclosure is ready in the Library makerspace, students proceed to the assembly phase to complete the product. Students then solder the required electronic components onto the PCB, including ESP8266 module headers, transistors, switches, and sensors, ensuring secure, reliable connections. After verifying the assembled board for proper functionality, the next step is to fit the PCB, 9V battery, and ESP8266 module into the custom 3D printed enclosure. Components are positioned according to the enclosure's mounting features, with wiring and connectors arranged for ease of access and neatness. The enclosure is then closed using snap fits or screws, securing all internal parts.

2.4 ESP8266 Programming

In the next workshop activity, students will program the ESP8266 module using basic C/C++ code provided through the Arduino framework. Students will begin by reviewing and analyzing the sample code to understand its structure and purpose. Next, they will customize the code to meet

Question #	Min	Mean	Max	SD
Q3	2	2.86	4	0.90
Q4	1	2.14	3	0.90
Q5	1	2.71	4	1.38
Q6	1	2.86	4	1.21
Q7	1	2.71	4	1.25
Q8	1	3.86	5	1.68

Table 1: Pre-workshop Quantitative Survey Analysis

the project's specific requirements and functionalities. The main objectives include initializing required variables, establishing a reliable internet connection, and configuring the system to continuously monitor the sensor output. If any changes are detected in the sensor readings, the code should immediately trigger an alarm and send a real-time notification to a mobile phone via the internet.

3 Pre- and Post-Workshop Survey

Survey data were collected from students using pre- and post-workshop questionnaires to assess their experiences and development. The pre-workshop survey captured baseline information, including year of study, prior exposure to product design, self-assessed confidence in CAD software, electronics, and manual assembly, as well as students' desired skill development. The post-workshop survey measured growth in specific competencies like 3D modeling, PCB layout, and soldering, and identified challenges faced during the process. It also assessed students' preparedness for future projects, satisfaction with the workshop instruction, and sense of belonging in their academic discipline. Anonymity was maintained, and participation was voluntary, allowing for meaningful analysis of confidence, skills, and attitudes influenced by the workshop.

The pre-workshop survey indicates varied student backgrounds as shown in Table 1. Confidence in using a soldering iron is highest (Mean = 3.86, SD = 1.68), while experience with electronics/circuit design is lowest (Mean = 2.14, SD = 0.90). Experience and confidence in CAD software and manual assembly are moderate but show considerable variability. These results highlight strong skills in soldering but identify electronics and circuit design as key areas needing support during the workshop.

The post-workshop survey results show consistently high levels of confidence and satisfaction among participants as seen in Table 2. Mean scores for all items range from 3.86 to 4.29, with relatively low standard deviations, indicating both strong and uniform gains. Confidence is highest for soldering without defects (Q12, Mean = 4.29), while ratings of technical demonstration clarity (Q15, Mean = 4.00, SD = 0.58) and other skills are similarly positive. These findings suggest that the workshop was effective in boosting students' technical skills and was well-received in terms of instructional quality.

In addition to quantitative survey questions, the post-workshop survey included items related to sense of belonging in the engineering community, measured on a Likert scale. The results shown in Figure 1 indicate that, after the workshop, most students feel positively about their engineering

5 Future Work

Future research could focus on evaluating the quality of products manufactured by students as an additional measure of the educational intervention's effectiveness. Since the workshop activity is currently offered as an extra-curricular experience rather than being integrated into the undergraduate engineering curriculum, assessing the finished products would provide further insight into the practical skills students acquire outside their formal coursework. Including product quality assessments alongside student feedback in subsequent studies could thus strengthen the overall evaluation of the intervention's impact.

References

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- [3] Cynthia J Finelli, Inger Bergom, and Vilma Mesa. Student teams in the engineering classroom and beyond: Setting up students for success. crlt occasional paper no. 29. *Center for Research on Learning and Teaching*, 2011.
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Appendix 1:

You do not have to participate in this research. Taking part in the research study is voluntary. Your decision to participate or to decline the research will not result in any penalty or loss of benefits to which you are entitled.

IMPLIED CONSENT TO TAKE PART IN RESEARCH

By completing this survey, I acknowledge that I am 18 years old and that I have read this consent form and that the research study has been explained to me. I agree to be in the research study described above. I may print a copy of this form for my records. By agreeing to participate, I have not given up any of the legal rights that I would have if I were not a participant in the study.

Pre-Workshop Survey

1. What is your current year of study?

- Freshman
- Sophomore
- Junior
- Senior

2. Have you ever designed or manufactured a physical product before?

- Yes
- No

3. Rate your previous experience with CAD Software (SolidWorks, Fusion360, etc.)

- 1 (No Experience)
- 2
- 3
- 4
- 5 (Expert)

4. Rate your previous experience with Electronics/Circuit Design

- 1 (No Experience)
- 2
- 3
- 4
- 5 (Expert)

5. Rate your previous experience with Manual Assembly (Soldering, tools)

- 1 (No Experience)
- 2
- 3
- 4
- 5 (Expert)

6. How confident are you in navigating the SolidWorks interface?

- 1 (No Experience)
- 2
- 3
- 4
- 5 (Expert)

7. How confident are you in reading a circuit schematic and translating it to a PCB layout?

- 1 (No Experience)
- 2
- 3
- 4
- 5 (Expert)

8. How confident are you in using a soldering iron safely?

- 1 (No Experience)
- 2
- 3
- 4
- 5 (Expert)

9. What is the #1 skill you hope to master by the end of this workshop?

Post-Workshop Survey

10. How confident are you now in creating a 3D enclosure in SolidWorks that fits a specific PCB footprint?

- 1 (Not confident)
- 2
- 3
- 4
- 5 (Very confident)

11. How confident are you now in routing traces on a PCB while following design rules?

- 1 (Not confident)
- 2
- 3
- 4
- 5 (Very confident)

12. How confident are you now in soldering components without "bridges" or "cold joints"?

- 1 (Not confident)
- 2
- 3
- 4
- 5 (Very confident)

13. Which part of the design process did you find the most challenging?

- 3D Modeling (SolidWorks)
- PCB Layout
- Physical Soldering/Assembly

14. Do you feel more prepared to tackle a personal or capstone project after this workshop?

- Yes
- No

15. How would you rate the clarity of the technical demonstrations?

- 1 (Poor)
- 2
- 3
- 4
- 5 (Excellent)

16. What was your "lightbulb moment" during the workshop?

17. Is there any specific tool or technique you wish we had spent more time on?

18. Overall, I am happy with the major I've chosen.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

19. I don't intend to change my major from my current major to another major.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

20. I am enthusiastic about my major

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

21. I think I will be very happy to spend the rest of my career in my current major.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

22. My major is interesting to me.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

23. I do not feel like "part of the family" in my academic discipline.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

24. I do not feel "emotionally attached" to my academic discipline.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

25. I do not feel a strong sense of "belonging" to my academic discipline.

- Strongly Disagree
- Disagree
- Neutral
- Agree
- Strongly Agree

Appendix 2:

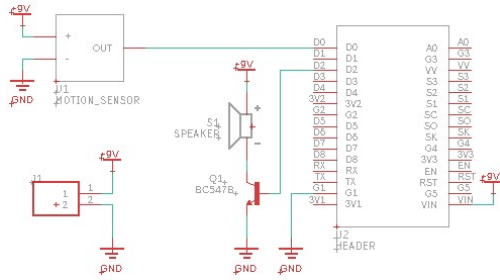


Figure 2: PCB Schematic

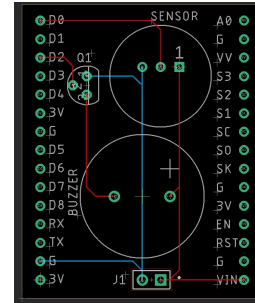


Figure 3: PCB Board Layout



Figure 4: Pre-assembled Product



Figure 5: PCB fit into 3D printed enclosure



Figure 6: Complete Product